PAPER ID-410082

Subject Code: KOE097

Roll No: | | | | | | | | | | | | | |

BTECH

(SEM VIII) THEORY EXAMINATION 2023-24

BIG DATA

TIME: 3 HRS **M.MARKS: 100**

Note: 1. Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt *all* questions in brief. **2 x 10 = 20**

Q no.	Question	Marks	CO
a.	List any five Big Data platforms.	02	1
b.	Discus the importance of Hadoop technology in Big Data Analytics.	02	1
c.	Explain three benefits of MapReduce.	02	2
d.	Define heartbeat in HDFS.	02	2
e.	List any five Big Data platforms.	02	3
f.	Define data replication in Hadoop Distributed File System.	02	3
g.	Name any two data ingestion tool in Hadoop.	02	4
h.	Compare and Contrast No SQL and Relational Databases	02	4
i.	Discuss the advantages of scala over java.	02	5
j.	Differentiate between Pig and Hive	02	5

SECTION B

2. Attempt any *three* of the following: **3 x 10 = 30**

a.	Differentiate between structured, semi-structured and unstructured data with suitable example.	10	1
b.	Illustrate the anatomy of a MapReduce job run.	10	2
c.	Demonstrate the design of HDFS and concept in detail.	10	3
d.	Explain how CRUD operations with example are performed in MongoDB.	10	4
e.	Illustrate how Zookeeper facilitates coordination and synchronization among HBase.	10	5

SECTION C

3. Attempt any *one* part of the following: **1 x 10 = 10**

a.	Discuss the 5 Vs of Big Data and their implications.	10	1
b.	Elaborate various components of Big Data architecture.	10	1

4. Attempt any *one* part of the following: **1 x 10 = 10**

a.	Explain the architecture of Hadoop Distributed File System (HDFS) and its fault tolerance mechanisms.	10	2
b.	Discuss Hadoop streaming and pipes.	10	2

5. Attempt any *one* part of the following: **1 x 10 = 10**

a.	Examine how a client read and write data in HDFS.	10	3
b.	Mention about the Cluster specification? Describe how to Setting up a Hadoop Cluster?	10	3

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6. Attempt any *one* part of the following: 1 x 10 = 10

a.	State features of Apache Spark and also explain three ways of how Spark can be built with Hadoop components.	10	4
b.	Discuss about Name node high availability and HDFS federation.	10	4

7. Attempt any *one* part of the following: 1 x 10 = 10

a.	Discuss the need for PIG in Hadoop ecosystem. Also explain its various execution mode.	10	5
b.	Explain Apache Hive architecture and the components involved in its operation.	10	5

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